Efficient Quantum Chemistry at Scale: Generative ML, Sample Based Quantum Diagonalization, and Learned Basis Transfer

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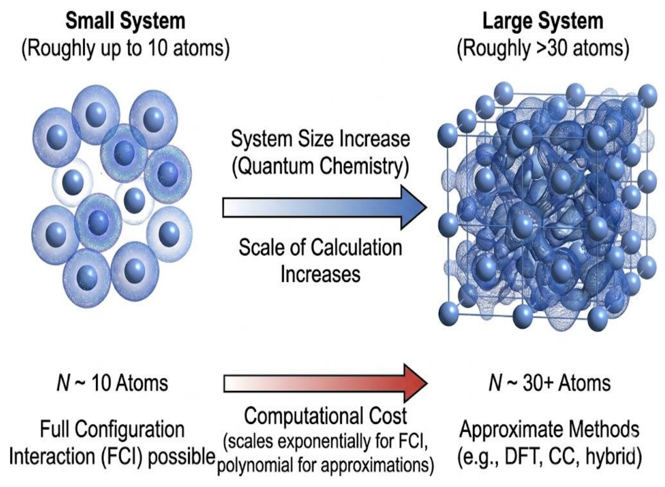
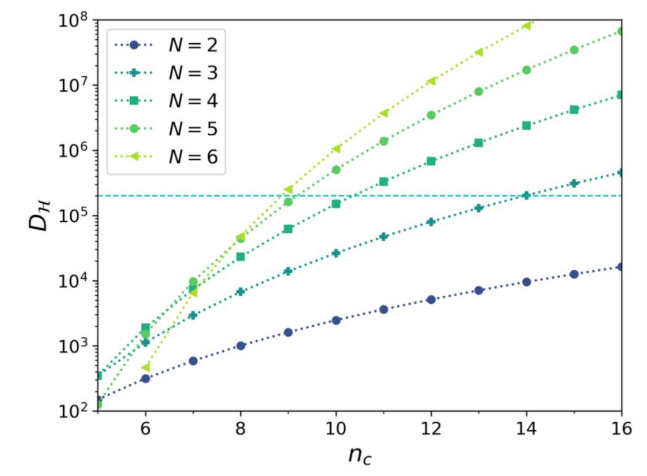
1. MOTIVATION

Many-Body Quantum Systems suffer from exponential Hilbert space growth



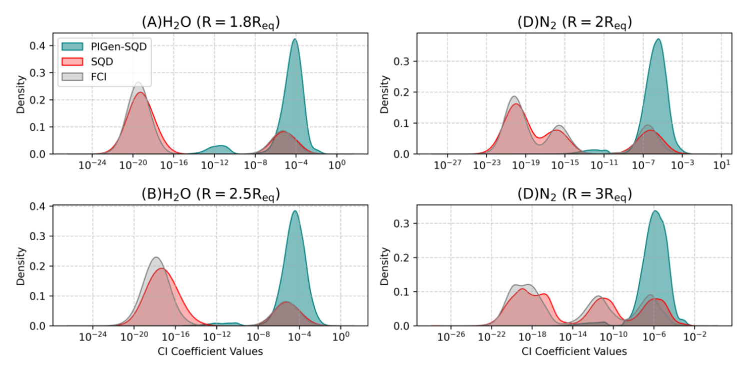


Exact methods like Full Configuration Interaction (FCI) become infeasible!

(Out of Memory OOM!)

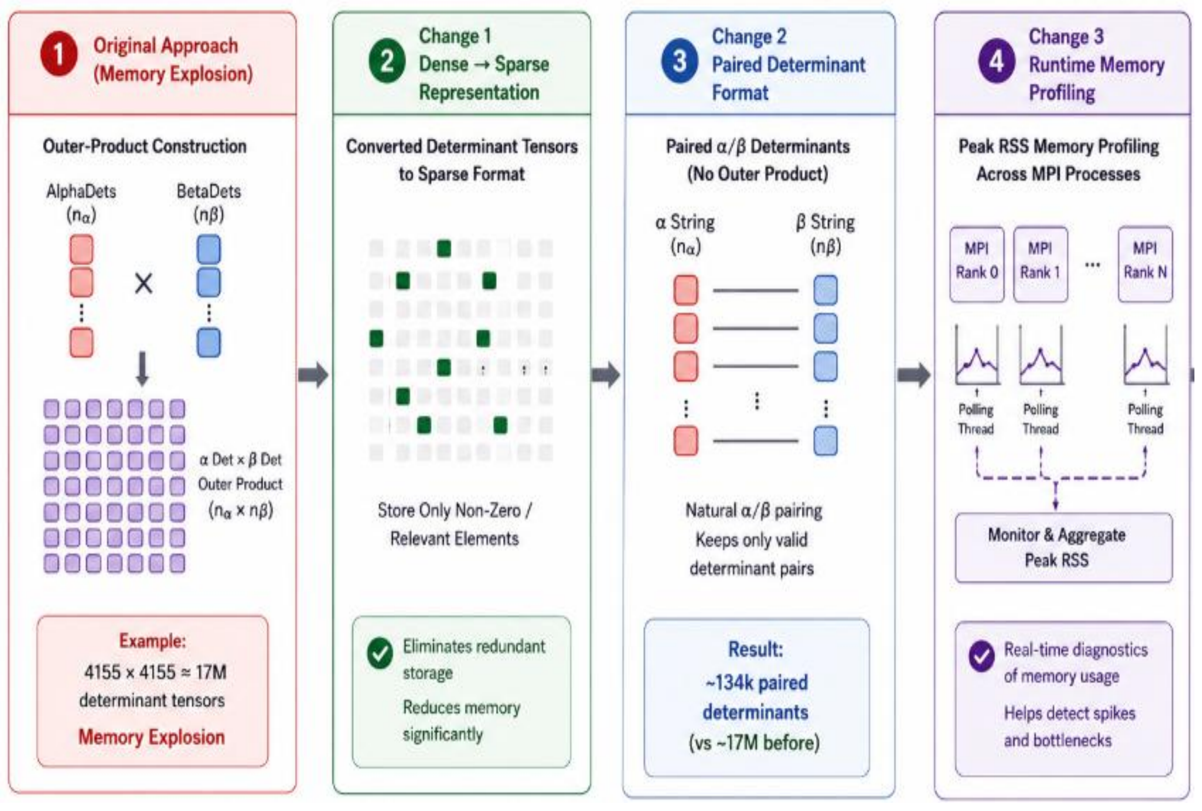
Need Scalable, Efficient and Transferrable Approaches

3. EFFICIENT DIAGONALIZER

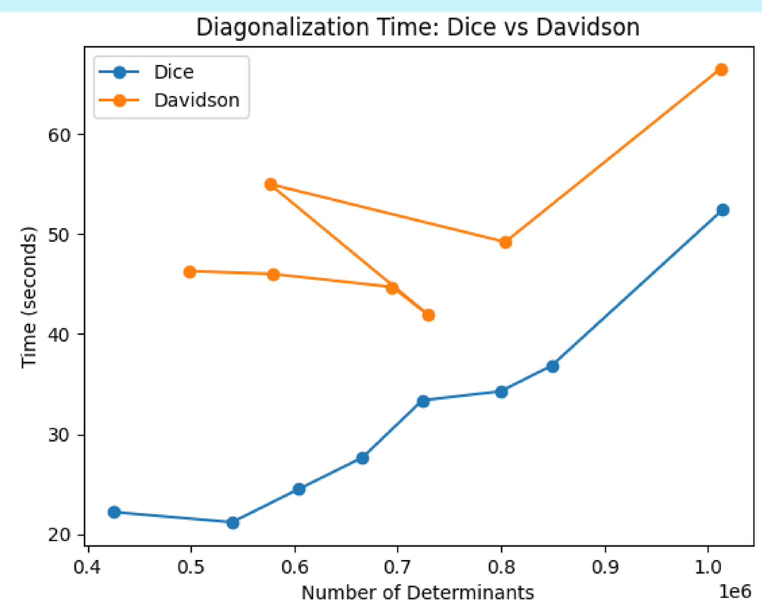


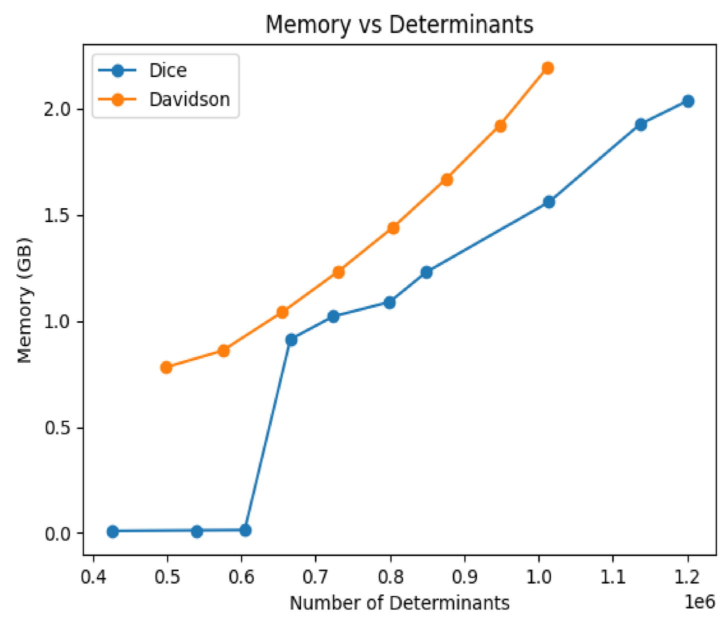
Only a small fraction of the configurations are actually important

Changes Done

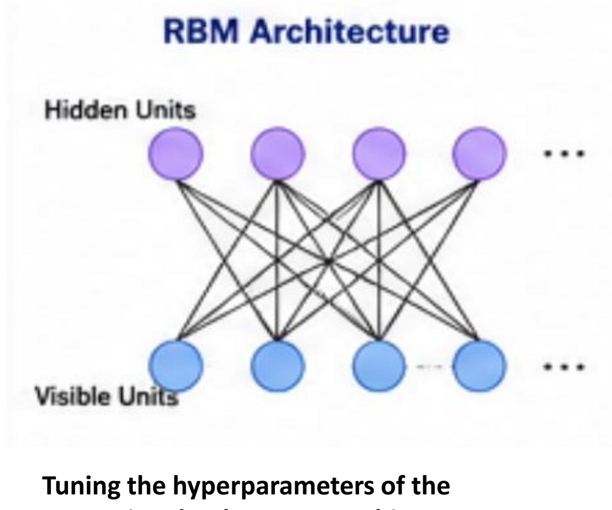


Linear Scaling Achieved!





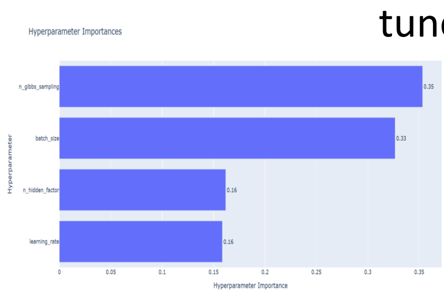
5. RBM HYPERPARAMETER OPTIMIZATION



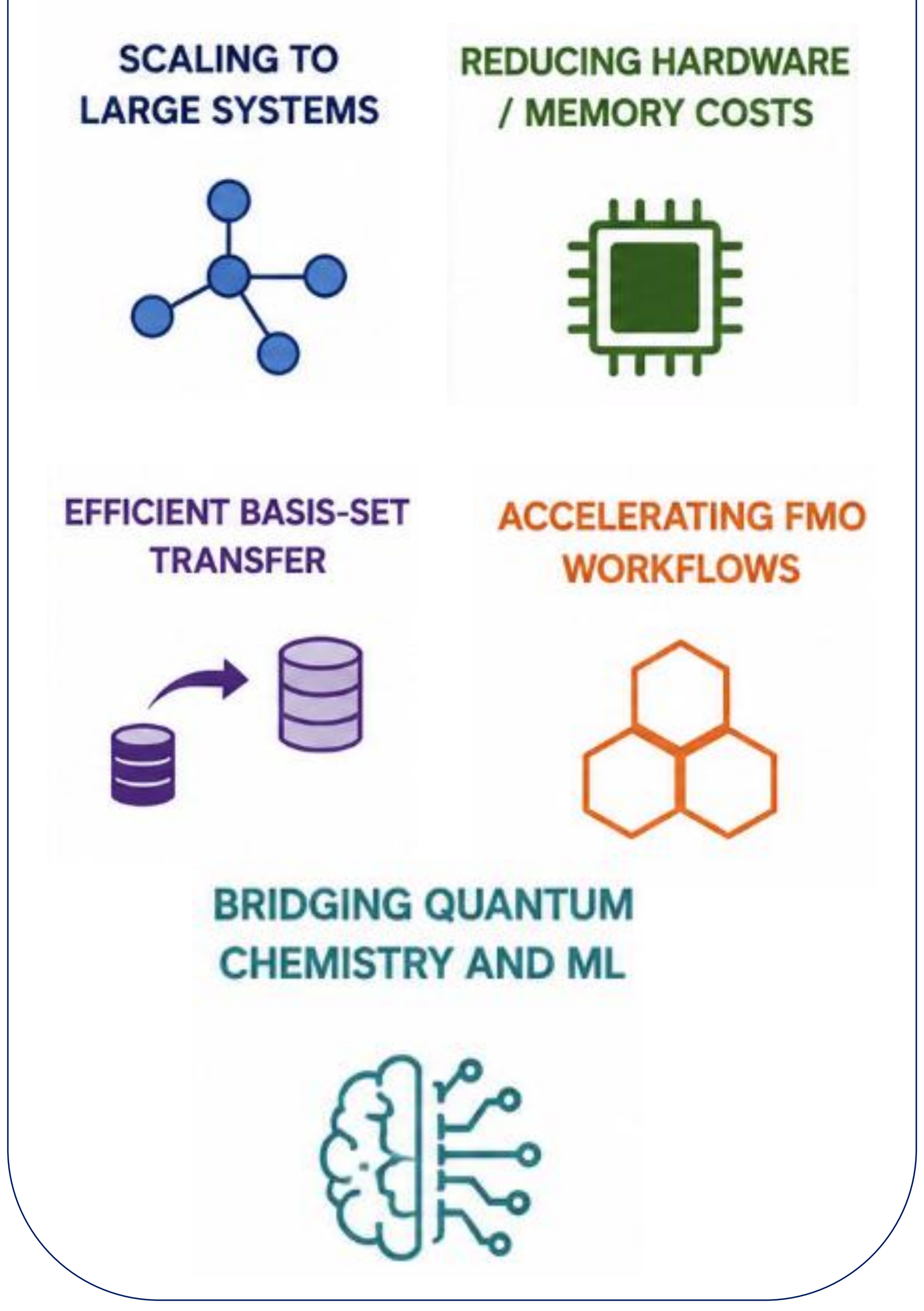
Tuning the hyperparameters of the Restricted Boltzmann Machine

Results:
Achieved 10e-4 Ha error with ~ 44% lesser configurations for H_2O

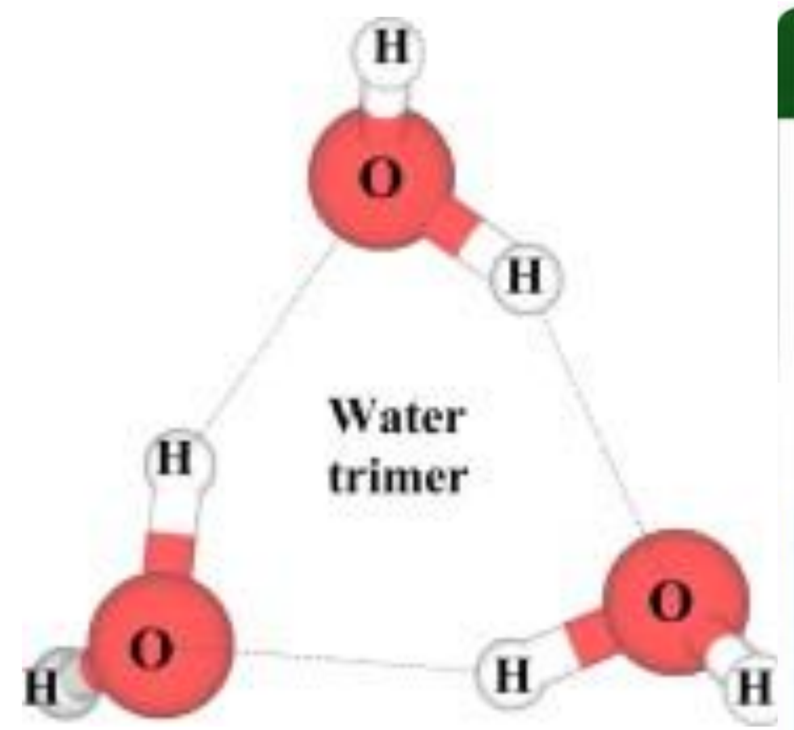
Optuna Hyperparameter tuner

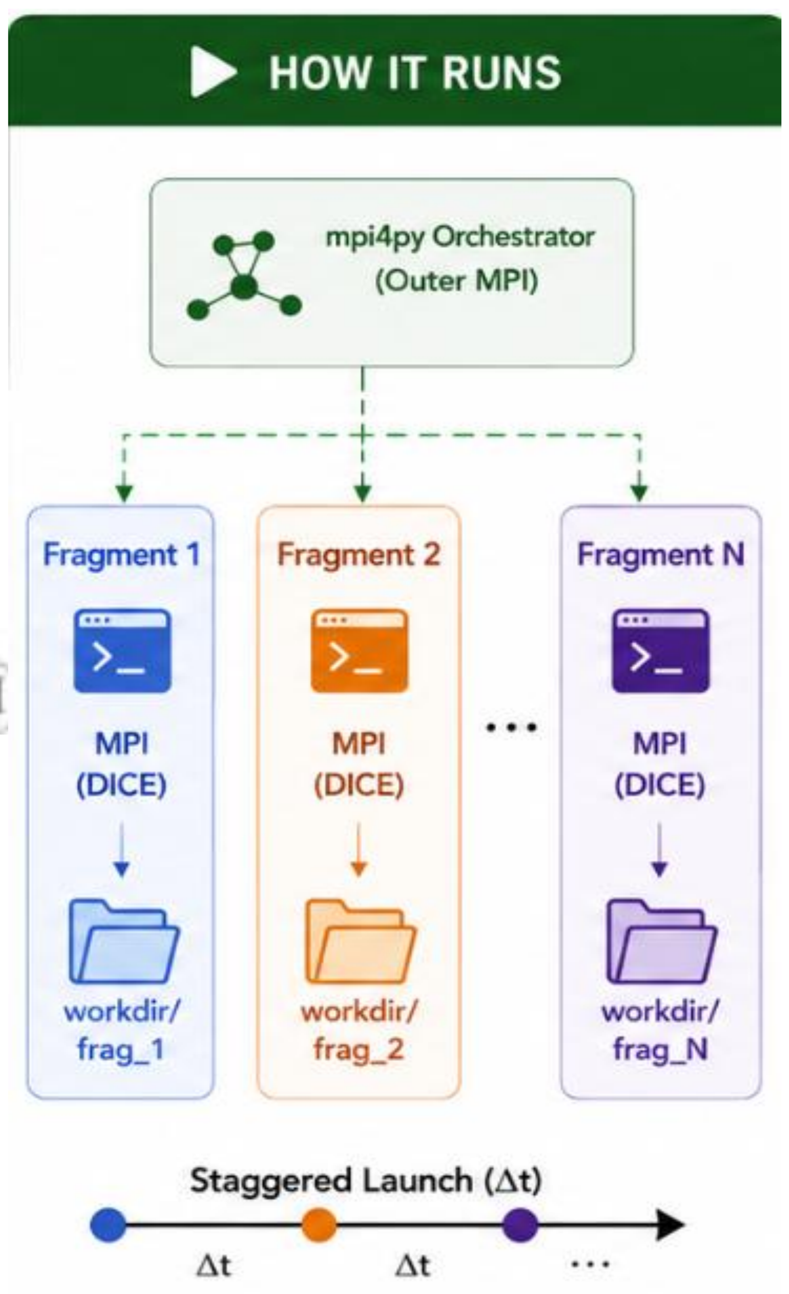


2. OBJECTIVES



4. PARALLEL FRAGMENT SOLVER*



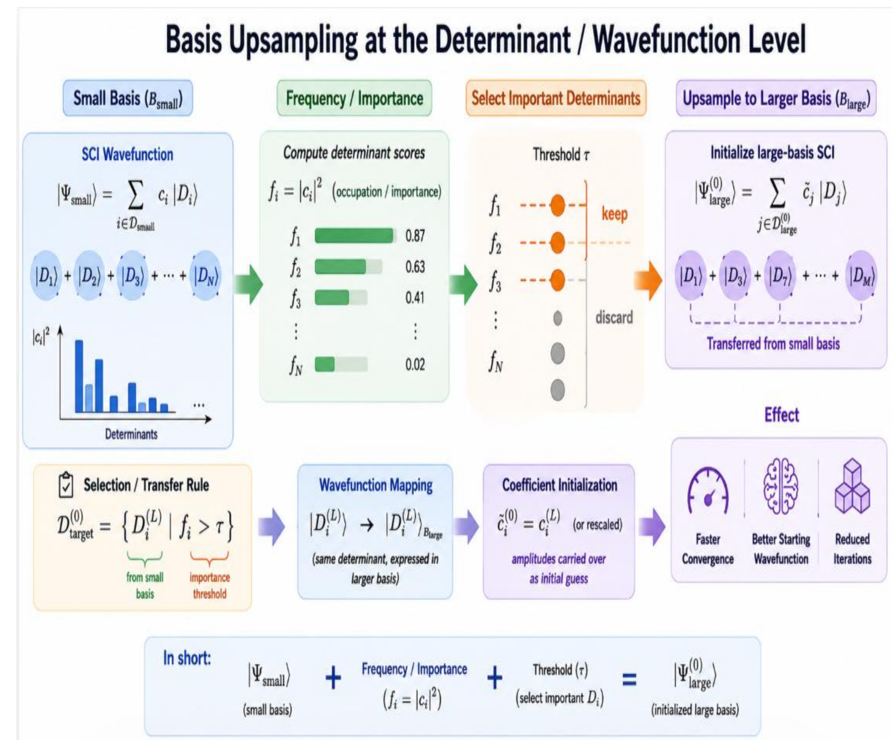


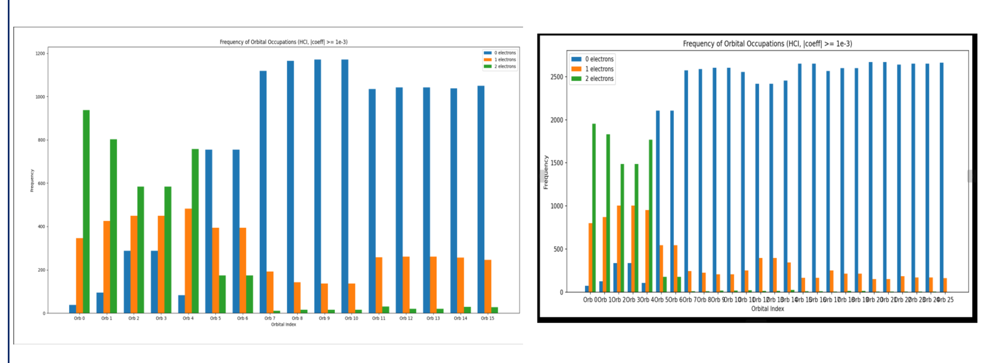
Used the modified dice solver and developed core level parallelization techniques to calculate energies for polymeric systems.

Solved Process-level parallelism over MPI-unsafe concurrent execution

*** Under publishing**

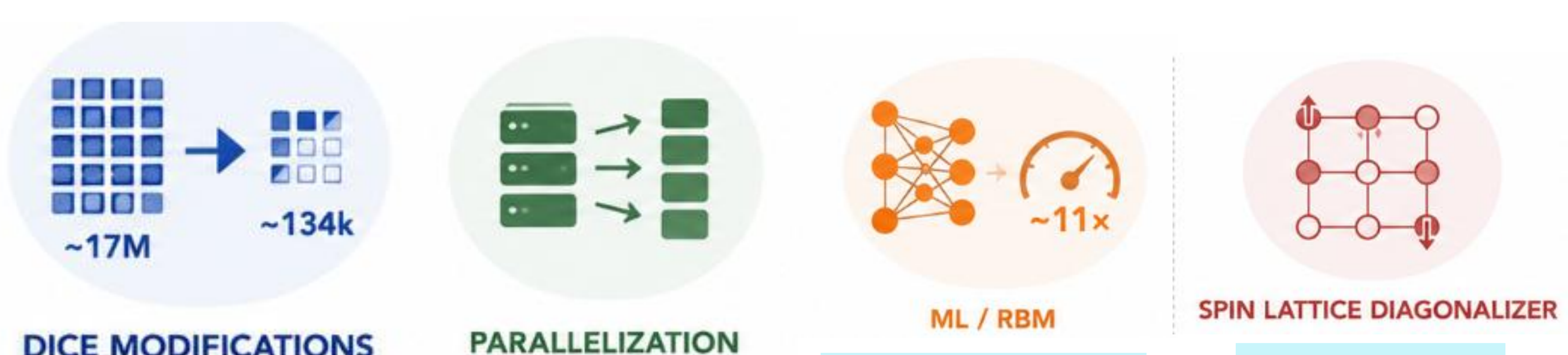
6. BASIS TRANSFER(ONGOING)





Qualitative Similarity of Configurations in the small and large basis

7. KEY OUTCOMES



Efficient Diagonalizer

Core Level FMO parallelization

Speedup and improved ML workflow

DICE Inspired Solver for Spin Chain Problem

8. FUTURE DIRECTIONS

- ➔ Structured Generation of Dominant Configurations
- ➔ Extension to Larger Active Space
- ➔ Scalable Parallel and low cost algorithmic developments
- ➔ Adaptive Transfer of Configurations across geometries and Molecules
- ➔ Better GenML models

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